

Day 14 – Armstrong Number (Part 1)

Aaj ka topic:

Armstrong Number

Ye **Basic Mathematics** ka bahut important interview question hai. Isse tumhari **digit extraction, power calculation**, aur **loop logic** strong hoti hai.

Problem Statement

Ek integer n diya hai.

Check karna hai ki wo **Armstrong Number** hai ya nahi.

Armstrong Number Kya Hota Hai?

Definition:

Har digit ki power (total digits jitni) ka sum agar original number ke equal ho, to us number ko Armstrong Number kehte hain.

Ye definition pehli baar me thodi confusing lag sakti hai.

Isliye example se samajhte hain.

Example 1

Input

153

Sabse pehle digits count karo.

153

↓

3 digits

Ab har digit ki power 3 hogi.

$$1^3 = 1$$

$$5^3 = 125$$

$$3^3 = 27$$

Ab sum karo.

$$1 + 125 + 27$$

=

$$153$$

Ye original number ke equal hai.

Isliye

$$153$$

↓

Armstrong Number

Example 2

Input

$$123$$

Digits

$$3$$

Power

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

Sum

$$1 + 8 + 27$$

=

$$36$$

Compare

36 != 123

Answer

Not Armstrong

Example 3

Input

9474

Digits

4

Ab har digit ki power 4 hogi.

$$9^4 = 6561$$

$$4^4 = 256$$

$$7^4 = 2401$$

$$4^4 = 256$$

Sum

$$6561 + 256 + 2401 + 256$$

=

9474

Answer

Armstrong

Question ko Kaise Samjhen?

Interview me agar question aaye:

Check whether a number is Armstrong or not.

Tumhare dimaag me ye steps aane chahiye.

Step 1

Digits Count Karo

↓

Step 2

Har digit nikalo

↓

Step 3

Us digit ki power = total digits

↓

Step 4

Sab powers ko add karo

↓

Step 5

Original number se compare karo

Bas.

Kitni Approaches Hain?

Is problem ke liye **sirf ek standard practical approach** hai.

☐ Mathematical Approach

Kyun?

- Interview me isi ka use hota hai.
 - Competitive Programming me bhi isi ka use hota hai.
 - String approach unnecessary hai.
-

Algorithm

Suppose

`n = 153`

Step 1

Digits Count

3

Step 2

Original Number Save

`original = n;`

Kyun?

Kyuki baad me

`n = n / 10;`

karte karte `n = 0` ho jayega.

Step 3

Har digit nikalo.

`digit = n % 10;`

Step 4

Power nikalo.

```
Math.pow(digit, digits)
```

Step 5

Sum me add karo.

Step 6

Last digit hata do.

```
n = n / 10;
```

Step 7

Loop khatam hone ke baad compare karo.

```
sum == original
```

Java Code

```
import java.util.Scanner;

public class ArmstrongNumber {

    public static int countDigits(int n) {

        if (n == 0)
            return 1;

        int count = 0;

        while (n != 0) {
            count++;
            n = n / 10;
        }

        return count;
    }

    public static boolean isArmstrong(int n) {

        int original = n;

        int digits = countDigits(n);

        int sum = 0;

        while (n != 0) {

            int digit = n % 10;

            sum += Math.pow(digit, digits);

            n = n / 10;
        }

        return sum == original;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number : ");

        int n = sc.nextInt();

        if (isArmstrong(n))
            System.out.println("Armstrong Number");
        else
            System.out.println("Not Armstrong");
    }
}
```

Line by Line Explanation

Line

```
int original = n;
```

Original number save kar liya.

Example

153

Loop ke baad

n

↓

0

ho jayega.

Isliye compare karne ke liye original chahiye.

Line

```
int digits = countDigits(n);
```

Pehle digits count karni padengi.

Example

153

↓

3

Kyuki har digit ki power 3 hogi.

Line

```
int sum = 0;
```

Power ka total sum store karega.

Line

```
while(n != 0)
```

Har digit process hogi.

Line

```
int digit = n % 10;
```

Last digit nikalo.

Example

153

↓

3

Sabse Important Line

```
sum += Math.pow(digit, digits);
```

Example

```
digit = 5
```

```
digits = 3
```

Calculation

5^3

=

125

Sum

1 +125 +27

Line

`n = n /10;`

Last digit hata do.

Final Line

`return sum == original;`

Agar

`153 ==153`

True.

Nahi to false.

Complete Dry Run

Input

153

Digits

3

Initially

`sum =0`

Iteration 1

Digit

3

Power

3^3

=

27

Sum

27

Remaining

15

Iteration 2

Digit

5

Power

125

Sum

$27+125$

=

152

Remaining

1

Iteration 3

Digit

1

Power

1

Sum

153

Remaining

0

Loop stop.

Compare

$153 == 153$

Answer

Armstrong Number

Dry Run (Not Armstrong)

Input

123

Digits

3

Power Sum

1

+

8

+

27

=

36

Compare

$36 \neq 123$

Answer

Not Armstrong

Time Complexity

Is algorithm me 2 **loops** hain.

Loop 1

Digits count karta hai.

$O(\log N)$

Loop 2

Har digit ki power calculate karta hai.

Again

$O(\log N)$

Total

$O(\log N) + O(\log N)$

=

$O(\log N)$

Big-O me constant ignore hota hai.

Space Complexity

Hum use kar rahe hain.

- original
- digits
- sum
- digit

Bas.

Space Complexity

=

$O(1)$

Advantages

- Interview standard solution.
 - Easy logic.
 - Extra memory nahi.
 - Har positive integer ke liye kaam karta hai.
-

Disadvantages

- `Math.pow()` floating-point value return karta hai (`double`), isliye bahut bade numbers me precision ka concept aata hai.
 - Har digit ke liye power calculate karni padti hai.
-

Common Mistakes

Mistake 1

Digits count nahi karna.

Har digit ko cube (³) kar dena.

☐ Galat.

Power **total digits** ke equal hoti hai.

Mistake 2

Original save nahi karna.

Loop ke baad

n

↓

0

Ho jayega.

Compare kis se karoge?

Mistake 3

`digit ^ digits`

❑ Wrong.

Java me ^ power nahi hota.

Ye **Bitwise XOR Operator** hai.

Power ke liye:

`Math.pow()`

use karo.

Edge Cases

Input

0

Output

Armstrong

Kyunki

0^1

=

0

Input

5

Output

Armstrong

Kyunki

5^1

=

5

Input

9474

Output

Armstrong

Interview Me Kaunsi Approach Likhen?

☐ Mathematical Approach

Kyun?

- Sirf yehi standard solution hai.
- $O(\log N)$ Time.
- $O(1)$ Space.
- Interviews aur Competitive Programming dono me accepted.

🚀 Day 14 Part 2 me hum cover karenge:

- `Math.pow()` internally kya return karta hai?
- `Math.pow()` se related interview traps.
- `double` se `int` conversion.
- `Math.pow()` ke bina power kaise calculate karen (loop-based power function).
- Kaunsi approach interview me zyada pasand ki jati hai aur kyun.

next

Day 14 – Armstrong Number (Part 2)

Part 1 me humne `Math.pow()` use kiya tha.

Aaj hum usko detail me samjhenge.

Ye part interview ke liye important hai.

Sabse Pehle Ek Question

Humne likha tha:

```
sum += Math.pow(digit, digits);
```

Question:

Ye `Math.pow()` karta kya hai?

Math.pow() Kya Hai?

Syntax

```
Math.pow(base, power)
```

Yahan

- **base** = jis number ki power nikalni hai.
- **power** = kitni power.

Example

```
Math.pow(2, 3)
```

Matlab

2^3

=

8

Example

```
Math.pow(5, 4)
```

Matlab

5^4

=

625

Armstrong Example

Input

153

Digits

3

Digit

5

Power

`Math.pow(5, 3)`

Result

125

Isi value ko sum me add kar dete hain.

Important Point

Bahut students sochte hain

`Math.pow()`

int return karta hai.

❑ Galat.

Ye return karta hai

double

Example

```
double x = Math.pow(2, 3);
```

```
System.out.println(x);
```

Output

8.0

Notice

8.0

Ye integer nahi hai.

Ye

double

hai.

Fir Hamara Code Error Kyu Nahi Deta?

Humne likha tha

```
sum += Math.pow(digit,digits);
```

Question

sum

kis type ka hai?

```
int sum = 0;
```

Aur

```
Math.pow()
```

return karta hai

double

Fir error kyu nahi?

Answer

Java me

`+=`

automatic type conversion kar deta hai.

Ye internally almost aise behave karta hai:

```
sum = (int)(sum + Math.pow(digit,digits));
```

Lekin agar tum ye likho:

```
sum = sum + Math.pow(digit,digits);
```

To compile error aayega, kyunki `sum + Math.pow(...)` ka result `double` hota hai aur usse bina cast ke `int` me store nahi kar sakte.

Isliye beginners ke liye ye version aur clear hai:

```
sum += (int) Math.pow(digit, digits);
```

Ya

```
sum = sum + (int) Math.pow(digit, digits);
```

Dry Run

Input

153

Digit

5

Calculation

```
Math.pow(5,3)
```

Return

125.0

Cast

```
(int)125.0
```

↓

125

Sum

27+125

=

152

Interview Question

Interviewer

Why did you use `(int)` before `Math.pow()`?

Answer

Because `Math.pow()` returns a double, but `sum` is an `int`. So we convert the result to an integer before adding it.

Kya `Math.pow()` Best Hai?

Answer

Mostly Yes

Lekin...

Interviewer kabhi kabhi bol sakta hai

Don't use built-in functions.

Tab?

Power Khud Calculate Karni Hogi

Example

5^3

Matlab

$5 \times 5 \times 5$

Custom Power Function

```
public static int power(int base, int exponent) {  
    int result = 1;  
    for (int i = 1; i <= exponent; i++) {  
        result = result * base;  
    }  
    return result;  
}
```

Line by Line

Line

```
int result = 1;
```

Answer yahan store hoga.

Loop

```
for(int i=1;i<=exponent;i++)
```

Jitni power hai utni baar multiplication.

Line

```
result = result * base;
```

Example

base =5

power =3

Initially

result=1

Iteration 1

1×5

=

5

Iteration 2

5×5

=

25

Iteration 3

25×5

=

125

Done.

Armstrong Using Custom Power

```
public static boolean isArmstrong(int n) {  
    int original = n;  
    int digits = countDigits(n);  
    int sum = 0;  
    while (n != 0) {  
        int digit = n % 10;  
        sum += power(digit, digits);  
        n = n / 10;  
    }  
    return sum == original;  
}
```

Bas ek line change hui.

Pehle

```
Math.pow(digit,digits)
```

tha.

Ab

```
power(digit,digits)
```

hai.

Time Complexity

Using Math.pow()

Digit loop

$O(\log N)$

Power calculation ko algorithmic analysis me constant maan lete hain.

Final

$O(\log N)$

Using Custom Power

Suppose

Digits

5

Har digit ke liye

5 multiplications.

Agar total digits bhi d hain, to overall roughly $d \times d$ operations ho sakte hain.

Isliye:

Time Complexity = $O(d^2)$

Aur kyunki $d = \log_{10} N$:

$O((\log N)^2)$

Practical note: Java `int` me maximum 10 digits hote hain, isliye ye bhi fast hi chalega, lekin theoretical analysis me $O((\log N)^2)$ maana jata hai.

Space Complexity

Dono methods me

$O(1)$

Comparison

Feature	Math.pow()	Custom Power
Easy	☐	Medium
Built-in	☐	☐
Interview	☐ (unless restricted)	☐ (when built-ins not allowed)
Space	$O(1)$	$O(1)$
Time	$O(\log N)$ overall	$O((\log N)^2)$ overall

Common Mistakes

Mistake 1

`digit ^ digits`

☐ Wrong

Java me `^` ka matlab **XOR** hai.

Power nahi.

Mistake 2

`sum = sum + Math.pow(...)`

Bina cast ke.

☐ Compile Error.

Correct

```
sum = sum + (int)Math.pow(...);
```

Ya

```
sum += (int)Math.pow(...);
```

Mistake 3

Power function me

```
result = base;
```

Se start karna.

☐ Wrong.

Always

```
result = 1;
```

Interview Me Kaunsi Approach Likhen?

Agar restriction nahi hai

☐ `Math.pow()` use kar sakte ho.

Code simple aur readable hota hai.

Agar interviewer bole:

"Do not use built-in functions."

Tab

☐ Apna `power()` function likho.

Ye dikhata hai ki tum sirf library par depend nahi ho.